Brazilian Journal of Physics (2023) 53:134
https://doi.org/10.1007/s13538-023-01347-0

SOCIEDADE BRASILEIRA DE FÍSICA

CONDENSED MATTER

Synthesis and Characterization of Bi_2CoCrO_6: A Double Perovskite Material With Promising Optical and Magnetic Properties

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Received: 9 May 2023 / Accepted: 12 July 2023 / Published online: 30 July 2023
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Abstract
This paper reports on the investigation of the structural, optical, and magnetic properties of a new double perovskite material Bi_2CoCrO_6 (BCCO), which was synthesized using a gel combustion route. The purity of the material and the formation of a tetragonal structure with space group I4/mmm (#139) were confirmed through Rietveld refinement of an X-ray diffraction pattern. The presence of metal–oxygen bonding was confirmed through Fourier transform infrared spectrum (FTIR) analysis of the material. The band gap of the material was found to be 2.95 eV through UV-Vis spectroscopy and Tauc plot analysis. The material was also found to exhibit ferromagnetic behavior with a saturation magnetization of 0.3768 emu/g at room temperature, and a ferromagnetic Curie temperature of around 450 K. These findings suggest that BCCO has potential for use in spintronics applications.

Keywords Double perovskites · Gel combustion · Band gap · Ferromagnetic · Super-exchange

1 Introduction

In recent years, metal oxides having excellent magnetic, ferroelectric, and dielectric properties at room temperature (RT) have gained huge attention in the advancement of new devices such as magnetic sensors, magnetic storage devices, and supercapacitors. Especially perovskites (ABO_3) with room temperature FM and FE properties have potential use in the development of high-density magnetic storage devices, such as hard disk drives and magnetic tapes. The rare-earth (R) cobaltites with perovskite structure ($RCoO_3$) are already proved to be promising candidates in the field of thermoelectric and sensor [1]. They also exhibit magnetic as well as electronic phase transitions which are associated with the changes in spin state of cobalt ions. The magnetic property of $RCoO_3$ depends on the spin state of Co^{3+} ions, which can be tuned by increasing temperature from the low spin state (S = 0) to the high spin state (S = 2). Similarly, the $RCrO_3$ compounds are widely studied for their catalytic activity, stability in high temperature, and multiferroic behavior. Recently, Yin et al. [2] report magnetic, magnetocaloric, and magneto-electric effects due to R-R and R-Cr magnetic interactions in the $RCrO_3$ (R = Dy, Nd, Er, Tb) perovskites synthesized via the flux method. Double perovskites have a crystal structure identical to perovskite and contain two different cations, and have also gained attention for their promising properties such as half-metallic (HM) [3–5], HM ferromagnet (HM-FM) [6, 7] half-metallic ferrimagnetic (HM-FiM) [8, 9], ferromagnetic [10, 11], and antiferromagnetic [12]. Their simple preparation methods and fascinating properties make them a suitable candidate for various applications [13–15]. The experimental studies on DP candidate $Bi_2CrB'O_6$ (B' = Fe, Ni, Mn, Cu) have created much interest due to multiple magnetic phase transitions and room temperature multiferroic properties [16–19]. It is also observed that the Bismuth-based cobaltites with DP structure $Bi_2CoMnO6$ are the budding candidates with a large FM Curie temperature and the co-existence of order parameters, such as multiferroic, magnetodielectric, magnetocapacitance, and magnetoresistance [20, 21]. More recently, Lin et al. [22] have theoretically investigated both the electronic and magnetic properties of series of DPs $Bi_2BB'O_6$ (B, B' = CoCr, CrNi, FeNi, FeZn, CrZn, and CoZn). They observed that the DP material Bi_2CoCrO_6 possesses FM half-metallic nature due to the double exchange interaction between the two B site cations. Currently, there is no record of any previous research conducted on the DP material

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Bi_2CoCrO_6. Therefore, the primary objective of this project is to synthesize the BCCO DP material and investigate its properties related to structure, optics, and magnetism.

2 Experimental Methods

The DP material BCCO was synthesized using the gel combustion route. In this method, metal nitrates $Bi(NO_3)_3.5H_2O$, $Co(No_3)_2{\cdot}6H_2O$, and $Cr(NO_3)_3{\cdot}9H_2O$ were dissolved in double-distilled water and mixed thoroughly to obtain a homogeneous solution. Citric acid was added to this solution in a weight ratio of 12 times the metal nitrates, and the mixture was heated to 240 °C. To attain a pH level of 9, ammonia was added dropwise to the solution. The mixture was continuously stirred at 70 °C on a hot plate to obtain a dark gray–colored resin. The resin was then heat-treated to decompose the nitrates and citrate completely, resulting in a black-colored powder. The powder was ground and calcinated at 600 °C for 1 h. The resultant powder was a brown-colored powder and was collected and preserved in a vacuum.

The phase and structure of BCCO were analyzed using powder-XRD (9 kW Rikagu, Japan) using Cu Kα radiation for the 2θ ranging from 10° to 90° in increments of 5°. The GSAS – EXPGUI software was used to perform the Rietveld refinement of the XRD pattern. A portion of the crystal structure of BCMO was obtained by using the 3-dimensional visualization system VESTA. UV–Vis spectroscopy was used as a tool to explore the optical properties of as-synthesized BCCO material. FTIR spectra of the DP material BCCO were recorded using a Spectrum Two FT-IR/Sp10 Software. Magnetic properties of the sample at room temperature (RT) were analyzed with the help of a vibrating sample magnetometer (VSM) (Lakeshore VSM 7410), and the high-temperature magnetic studies were done using VSM (Lakeshore 7410S).

3 Results and Discussions

The structural analysis of the material using XRD pattern is showed in Fig. 1a. It shows the presence of characteristic peaks of double perovskite systems. In Fig. 1a, a red-colored curve represents the observed data and a blue-colored line represents the pattern simulated by the Fullprof software. The black-colored base line is the difference between calculated and observed results. Rietveld refinement enabled us to identify that the material we synthesized crystallizes in a tetragonal structure with space group I4/mmm (#139). The calculated lattice parameters are $a = b = 5.4120$ Å and $c = 8.1312$ Å (Refer Table 1). These results are in good agreement with the theoretical values obtained from the Structure Prediction Diagnostic Software (SPuDS), which predicts that lattice constants are $a = 5.3968$ Å and $c = 8.1170$ Å for the BCCO material. This result suggests that our material does not have any impurity phase. Figure 1b shows the visualization of crystal morphology of our BCCO using VESTA (Visualization for Electronic and STructural Analysis) software. The average crystallite size of about 30 nm was calculated from the XRD pattern and Debye–Scherrer formula.

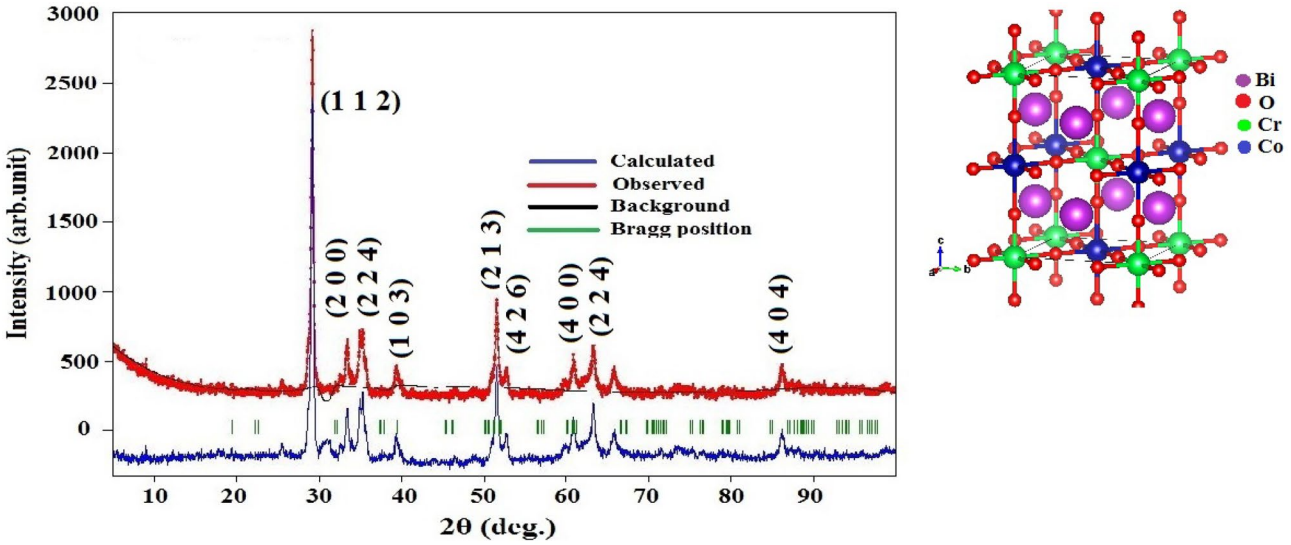


Fig. 1 **a** X-ray diffraction pattern of the as-synthesized Bi_2CoCrO_6 (red color) and the GSAS simulated pattern (blue color). **b** The structure of Bi_2CoCrO_6 using VESTA software (for clear understanding of the references to color in this figure legend, the reader is referred to the Web version of this article)

Table 1 Refined atomic parameters for Bi_2CoCrO_6 with the space group I4/mmm at RT

Atom	*x*	*y*	*z*
Bi1	0	0.5	0.25
Co1	0	0	0.5
Cr1	0	0	0
O1	0	0	0.235
O2	0.248	0.248	0

$a=5.4120$ A°, $b=5.4120$ A°, $c=8.1312$ A° $\alpha=\beta=\gamma=90°$

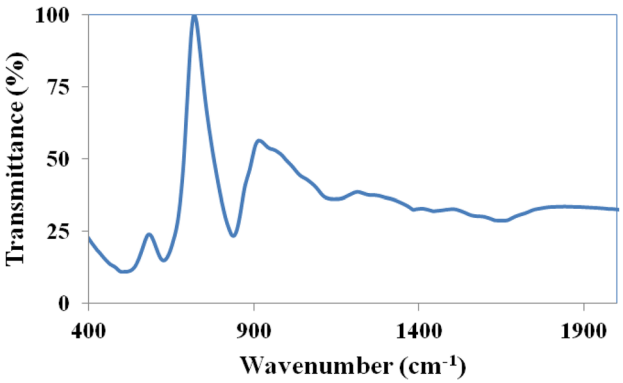


Fig. 2 FTIR spectra of Bi_2CoCrO_6 confirming the formation of double perovskites and absence of organic and inorganic impurities

The IR spectrum of BCCO has been recorded at room temperature (RT) over a range of wavenumbers from 300 to 1800 cm^{-1}. Figure 2 illustrates the IR spectrum showing several strong peaks that are characteristic of the Co-O and Cr-O stretching modes in the DP structure. Specifically, a strong peak at around 610 cm^{-1} is attributed to the stretching mode of Co-O [23], while a strong peak at 490 cm^{-1} is assigned to the Cr-O stretching mode. Additionally, a broad and strong peak centered around 843 cm^{-1} is due to the symmetric stretching vibration mode of Cr-O [24]. The IR spectrum confirms that the DP material we prepared is free from hydroxyl and carboxyl groups.

The X-ray photoelectron spectroscopy studies of Co 2p and Cr 2p are shown in Fig. 3a, b, respectively. The Co 2p XPS spectrum typically shows two peaks corresponding to the Co $2p_{3/2}$ and Co $2p_{1/2}$ core levels. The Co $2p_{3/2}$ peak is usually located at around 780–783 eV and the Co $2p_{1/2}$ peak is located at higher binding energy than the Co $2p_{3/2}$ peak, with a separation of about 14 eV between them. These peaks arise from the transitions of Co 2p electrons to unoccupied states, and their positions and intensities are sensitive to the local chemical environment of the cobalt atoms. The Co 2p peaks observed at 780.9 eV and 797.6 eV can indeed be attributed to the binding energies of Co^{2+} $2p_{3/2}$ and Co^{2+} $2p_{1/2}$, respectively, as these energies are consistent with those reported in the literature [25, 26]. The peaks present at 787.1, 805.0, 806.1, and 812.7 eV are the apparent shake-up satellite peaks. The positions and shapes of these peaks can provide information on the oxidation state and coordination geometry of the cobalt atoms in the sample. The Cr 2p spectrum (Fig. 3b) shows three peaks at 577.9, 579.2, and 587.4 eV, which can be assigned to the binding energies of Cr^{3+} $2p_{3/2}$, Cr^{6+} $2p_{3/2}$, and Cr^{3+} $2p_{1/2}$, respectively [27]. The percentage contributions of Cr^{3+} and Cr^{6+} to the total Cr content in the sample can be determined by measuring the areas under their corresponding peaks in the fitted curve. From the plot, it is observed that 71.68% of the Cr in the sample is in the +3 oxidation state, while 28.32% is in the +6 oxidation state. Similarly, the Co 2p spectrum shows two peaks, corresponding to Co^{2+} $2p_{3/2}$ and Co^{2+} $2p_{1/2}$ core levels. The contributions of these two states to the total Co content in the sample can be determined by

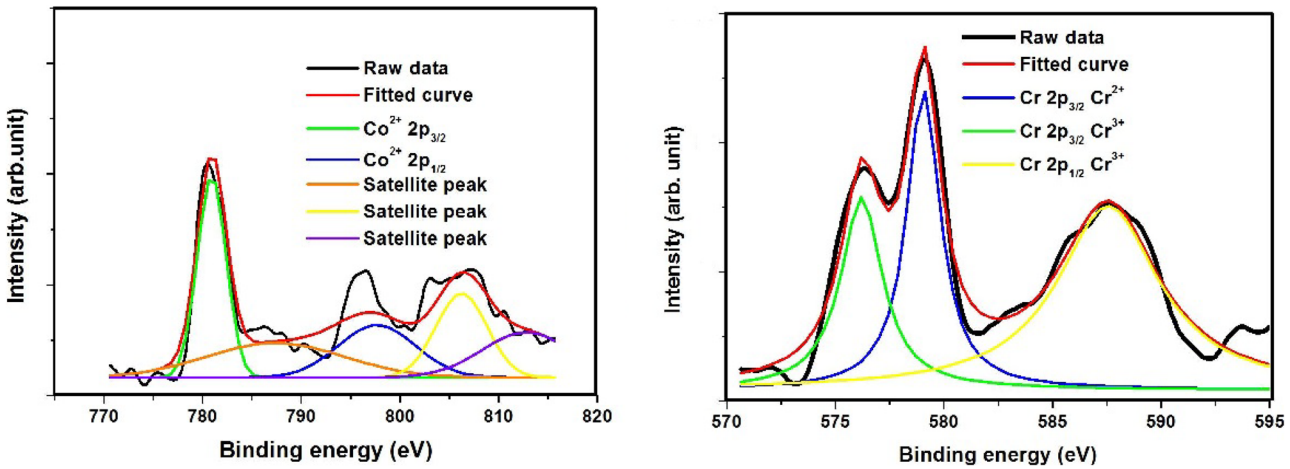


Fig. 3 **a** Co 2p XPS spectra of Bi_2CoCrO_6 and **b** Cr 2p XPS of Bi_2CoCrO_6 taken at 300 K along with fitted curves

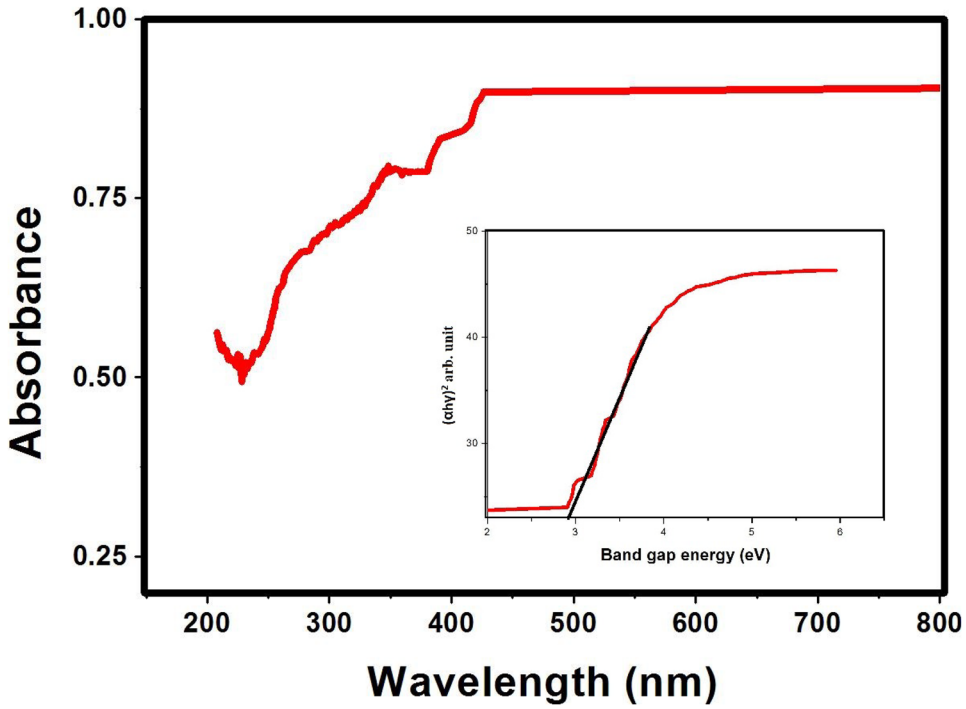


Fig. 4 UV−Vis absorption spectra for Bi_2CoCrO_6. The inset shows the corresponding Tauc plot for a direct band gap

measuring the areas under their corresponding peaks in the fitted curve. From the plot, 58.86% of the Co in the sample is in the + 2 $2p_{3/2}$ oxidation state, while 41.14% is in the +2 $2p_{1/2}$ oxidation state.

The UV-Vis absorption spectrum of the sample is recorded at RT and is illustrated in Fig. 4. The measurement has been made for the wavelength ranging from 200 to 800 nm. The Wood and Tauc formula relates the absorption coefficient (α) of a material to the photon energy ($h\nu$) and the band gap energy (Eg) through the following equation:

$$\alpha h\nu = A(h\nu - Eg)^n \quad (1)$$

where n is an exponent that depends on the nature of the electronic transitions, A is a constant related to the material's properties, and Eg is the band gap energy. The value of n is typically taken to be 2 for direct band gap materials and 1/2 for indirect band gap materials.

To estimate the value of direct optical bandgap energy of the sample, a plot was drawn between $(\alpha h\nu)^2$ and $h\nu$ and is illustrated in Fig. 4. Inset of Fig. 4 representing the Tauc plot exposed a direct band gap with energy of 2.95 eV. The above said energy was calculated by extrapolating the linear component of the curve to the energy axis ($(\alpha h\gamma)^2 = 0$). Such a wide band gap energy value of BCCO makes it a possible candidate for various applications, such as photovoltaic and optoelectronic devices [28, 29].

To determine the Curie temperature (TC) of our material, we have measured its magnetization as a function of temperature. The magnetization versus temperature plot (Fig. 5) shows a switch from PM to FM state as the temperature decreases. The Curie temperature is defined as the temperature at which the material undergoes this transition. In Fig. 5, it is observed that the TC of BCCO is 450 K, as determined from the peak value of the dM/dT plot (inset of Fig. 5). This value is higher than the TC of few double perovskites reported earlier, indicating that our material has a stronger FM character. To explore the magnetic properties of our material at RT, VSM has been used to measure the magnetic hysteresis loop of the material at 300 K. The results are displayed in Fig. 6. From the plot, it is observed that the material exhibits a magnetic hysteresis loop, indicating the presence of FM ordering. The saturation magnetization (Msat) of the material is found to be 0.3768 emu/g, indicating the strength of the FM interaction in the material. Additionally, the plot confirms that the material possesses a TC greater than 300 K. This indicates that the magnetic ordering in our material is more stable and robust at higher temperatures. The high TC value of the material suggests potential applications in magnetic storage and spintronic devices.

As in our DP, bismuth is non-magnetic, and magnetic properties of the sample depend on the B and B' site cations. In BCCO, the mixed oxidation states of Co and Cr in the B-site cations result in a variety of magnetic exchange and super-exchange (SE) mechanisms. The Co ion has a $3d^7$ electronic configuration and is in the Co^{2+} oxidation state, with a spin of S = 3/2. The Cr ion

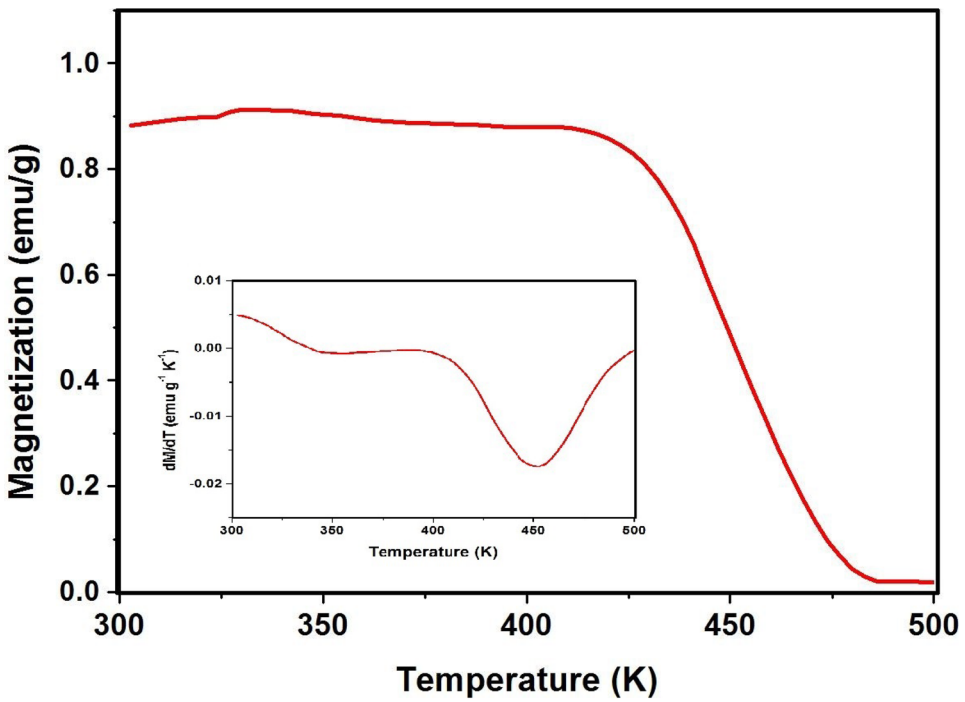


Fig. 5 Magnetization versus temperature curves at 1000 Oe showing a switch over from paramagnetic to ferromagnetic while cooling and inset shows Plot of dM/dT and temperature

exists in two oxidation states: Cr^{3+} with a $3d^3$ electronic configuration and a spin of S = 3/2, and Cr^{6+} with a $3d^0$ electronic configuration and a spin of S = 0. According to the Goodenough-Kanamori rule, the SE interactions lead to AFM ordering when the hopping of virtual electron occurs between overlapping orbitals that are each half-filled or having the same valence. Therefore, in our material, the SE mechanism occurs between Cr^{3+}-O-Cr^{3+}, Co^{2+}-O-Co^{2+}, and Cr^{6+}-O-Cr^{6+}, leading to AFM ordering. On the other hand, the direct exchange (DE) interaction occurs between Cr^{3+}-O-Cr^{6+} and Co^{2+}-O-Cr^{6+}, where the virtual electron transfer is between one half-filled and one

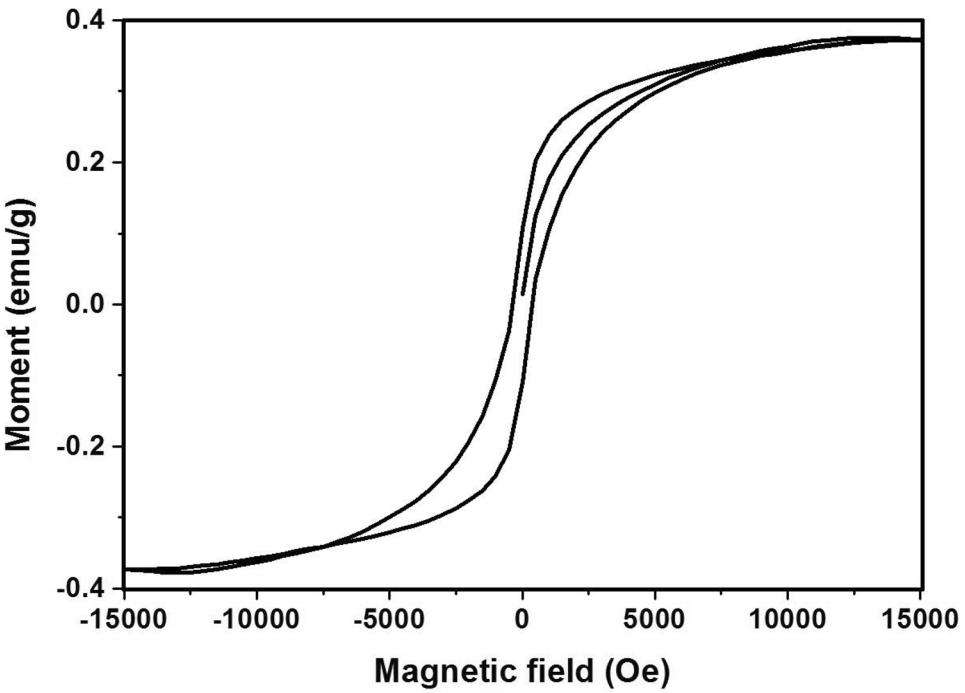


Fig. 6 M-H plot of Bi_2CoCrO_6 measured at RT showing FM ordering

empty orbital, leading to FM ordering. In the competition between the DE and SE mechanisms in BCCO, the FM ordering slightly dominates over the AFM ordering. However, the balance between these two mechanisms can be influenced by various factors such as the strength of the exchange interactions, the crystal structure, and the magnetic anisotropy. This FM behavior of BCCO matches with the theoretical investigation on the same DP material [22] in accordance with the GGA and GGA + U models.

4 Conclusion

We successfully synthesized a novel DP material called Bi_2CoCrO_6 using the gel combustion route. The material's crystal structure was determined to be I4/mmm through Rietveld refinement of powder X-ray diffraction data and all the peaks were indexed. Additionally, UV absorption spectrum measurements indicated that the material has a wider band gap energy. We also observed room temperature ferromagnetism in the material which could be attributed to the mixed valence of the B-site cations and the double exchange mechanism between the Cr t2g, O 2p, and Co t2g orbitals. Based on these findings, we suggest that the DP Bi_2CoCrO_6 material could be a promising candidate for spintronics applications due to its wider band gap energy and room temperature ferromagnetic behavior. Overall, our research contributes to the understanding of the properties and potential applications of novel DP materials.

Acknowledgements We acknowledge Nanotechnology Research Centre (NRC), SRMIST, SAIF, IIT, Chennai, Principal, Mepco Schlenk Engineering College, for providing the research facilities to complete this work.

Declarations

Competing Interests The authors declare no competing interests.

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